Finding scientific information

Research project

M12 BIT & TCS

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Information Specialist

EEMCS & LISA

Information Specialist?

Every faculty is supported by one or two information specialists.

We are library staff who help students and researchers with information literacy, access to books/journals/etc, publishing research, and more.

Always available to help, don’t hesitate to contact us! Information & contact info can be found on the library website, www.utwente.nl/library

Today's topic is information literacy

A 40-minute crash course to help you with:

selecting the right tools for the job

finding the sources you need

determining the value of your sources

effortlessly save and cite your sources

incorporating all of this in your research workflow

09:45 – 09:50	introduction, overview
09:50 – 10:05	1. tools & 2. workflow
10:05 – 10:15	3. complete process
10:15 – 10:30	4. wrap-up

The slides will be shared on canvas later today. They will include extra slides, links, etc.

For now: Keep your laptops and phones closed, focus on the talk: I'll be keeping the pace up! Raise your hand if you have a question.

Links to get you started

UT Library Service portal – UT's official list of databases, guides, contact, get access, software, and more

Information tools– my personal overview of useful tools, databases and websites – use options to filter & categorize

https://leanlibrary.com/download/ - browser plugin so you can automatically access articles even when outside UT network

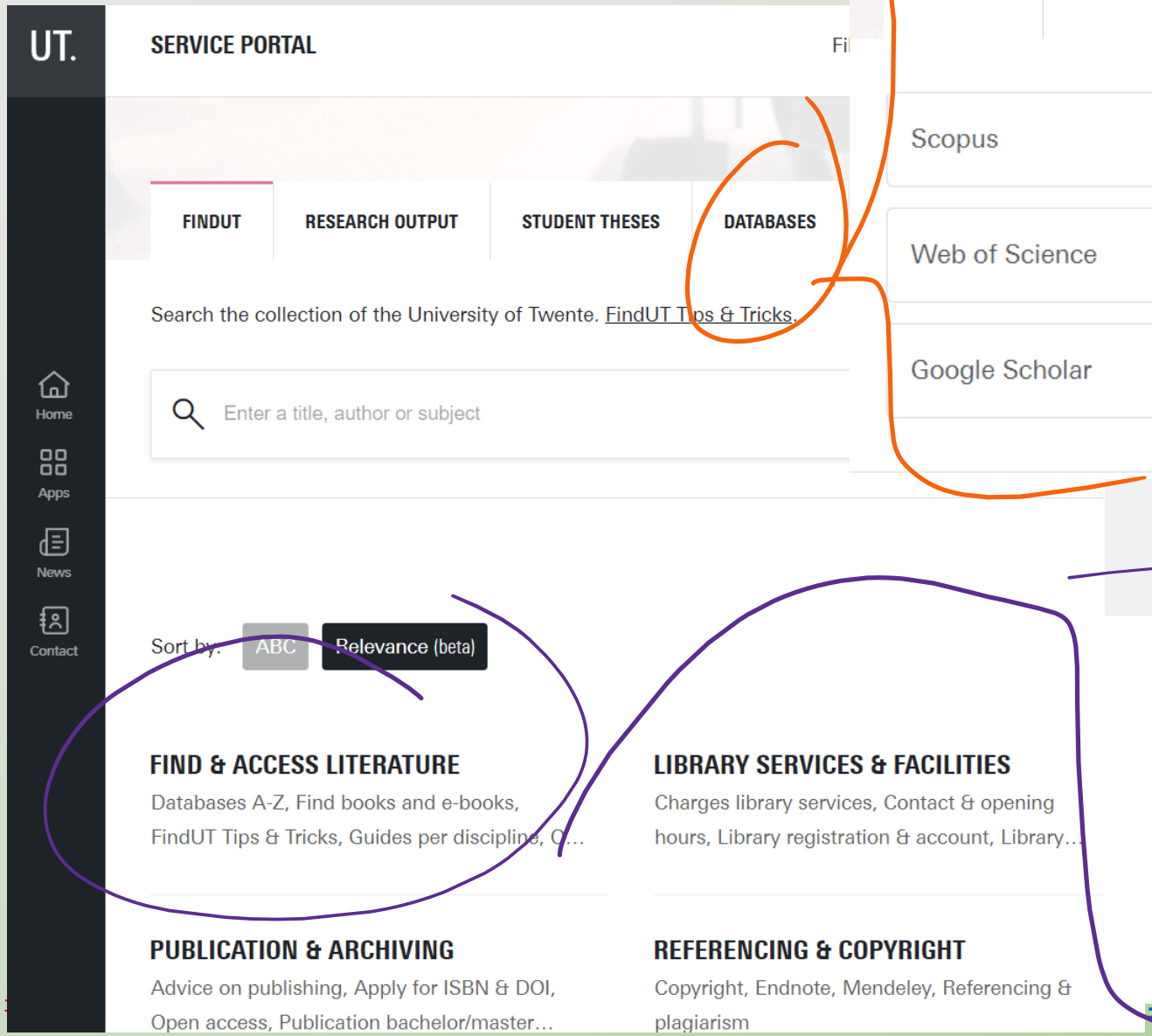
https://www.zotero.org/ & https://www.mendeley.com/ - ref managers, make sure to download browser plugin as well; come with Word plugin included

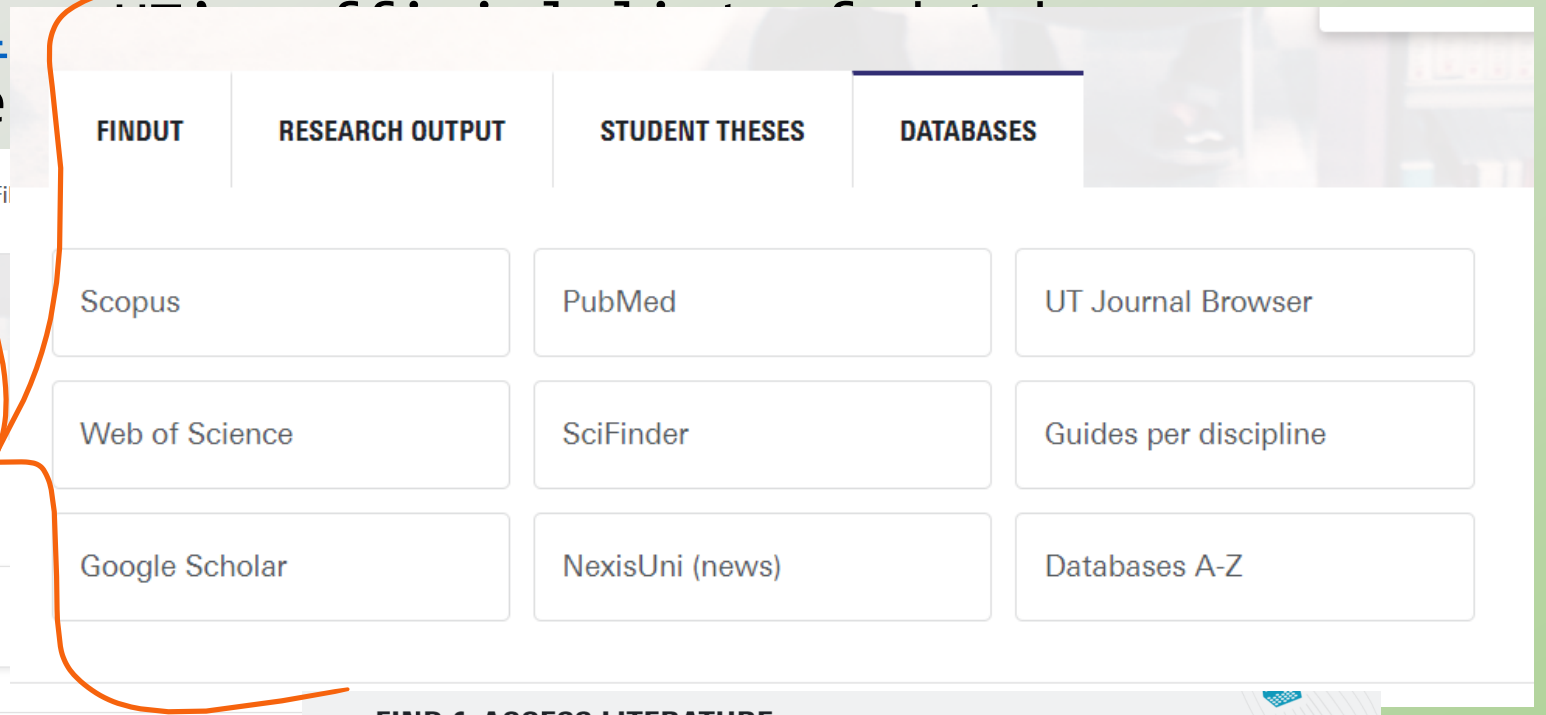
https://www.overleaf.com/learn/how-to/How to link your Overleaf account to Mendeley and Zotero

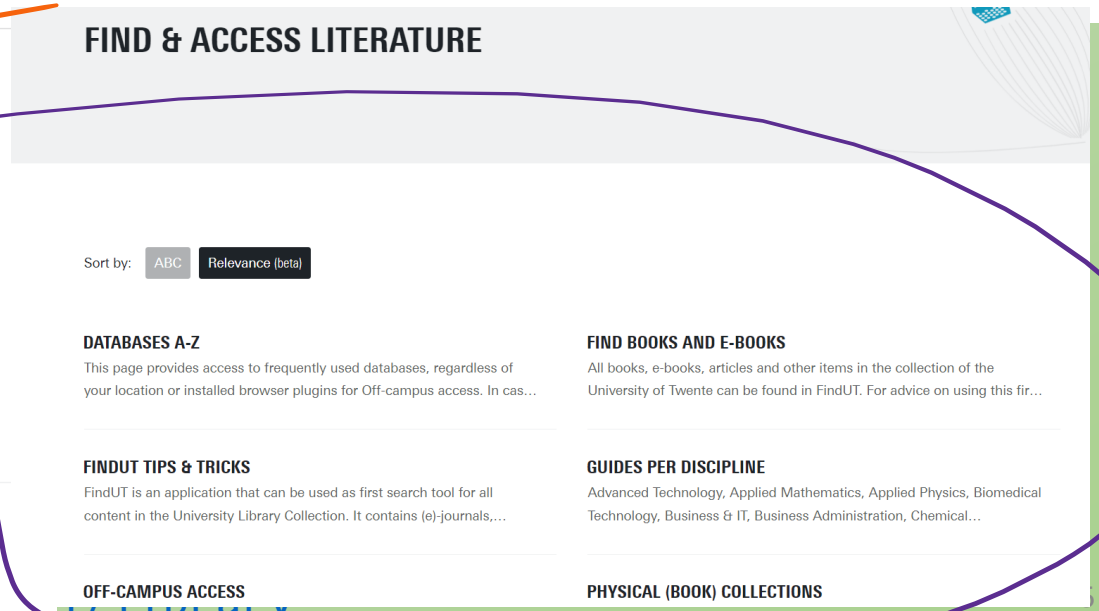
- Integrate zotero or Mendeley with Overleaf

UT Library Service portal

guides, contact, get acce







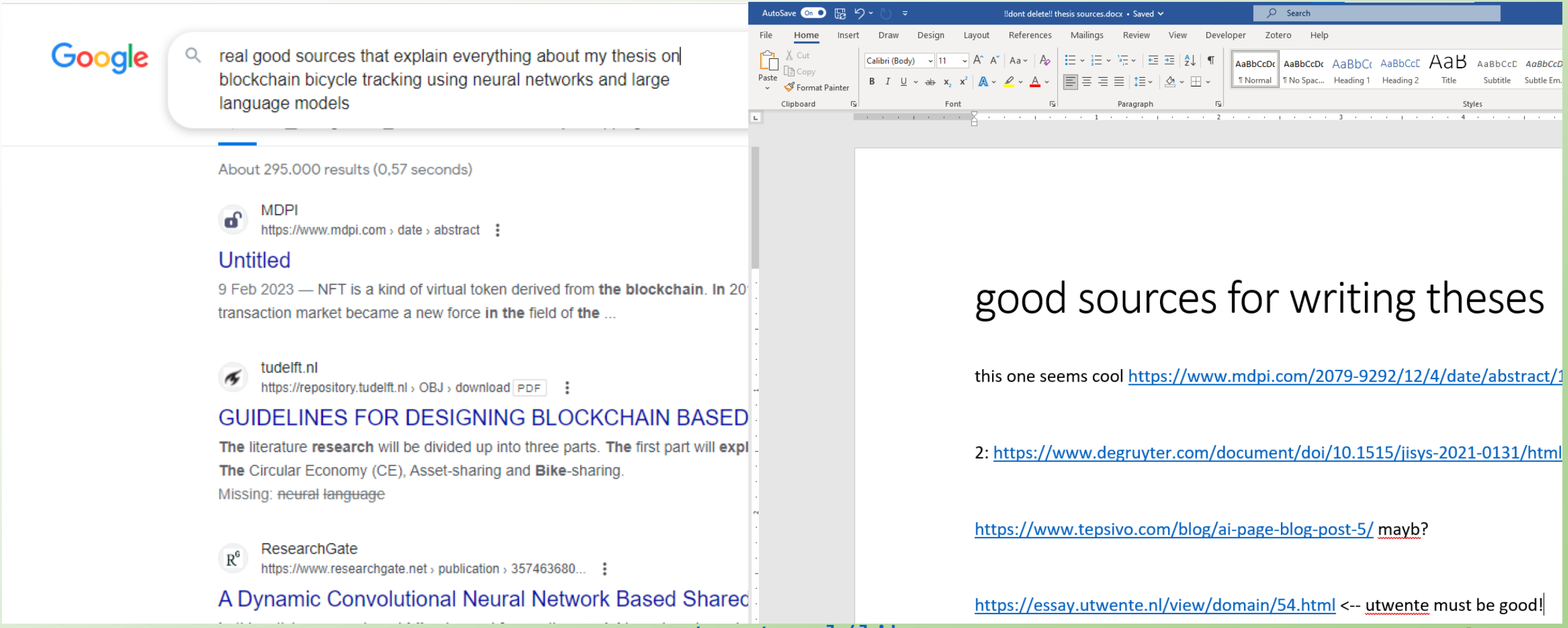
Information Literacy Tools

(no, google scholar is not all you need)

(using the proper tool for the job will save you a lot of time)

Information Literacy Tools

You need 2 things when doing a literature search: some way to find sources, and some way to store them. Here's an example:



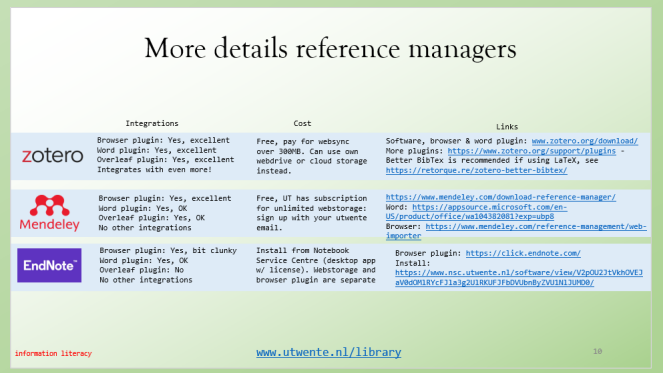
Install a reference manager & use it!

Please!!! use one. These three are the most used, but any that will let you store and cite with a single click will do. It will save you a lot of blood, sweat, and tears.
The three options listed below have browser plugins that enable single-click citations, and they also all integrate in Word (varying in ease of use).

	Description	Why pick this one?
zotero	My preferred manager. Free & open source. Can be customized to fit your needs.	- You want full control of the software & data and like playing around with settings - You don't mind a dated UI
Mendeley	Decent 2nd choice. By Elsevier (Scopus). Easy to set up and use, barely any customization.	- You want something that works out of the box - You don't have any niche requirements
EndNote™	Software with a long history. By Clarivate (Web of Science) Feels dated but is widely used.	- You want the true old-school experience - You want to use a product licensed by the UT

There are more reference managers out there, but most lack essential features compared to these three (like integration with text editors, no browser plugin, etc)

More details & links can be found here (click):



www.utwente.nl/library

Types of sources

Great: Peer-reviewed research & review articles (journals)

OK: Conference proceedings, patents, reports, guidelines, scientific books,

Be careful with: Government and company data/publications, newspapers, non-scientific journals

Don't recommend: Student theses

Only if *absolutely* necessary: General websites, social media, blogs, general books, something someone said once

Search tools

Keyword search

You probably know (some) of these:

Clarivate Web of Science™

Google Scholar

ELSEVIER Scopus® ScienceDirect

PubMed

IEEE Xplore®

siam Society for Industrial and Applied Mathematics | Publications Library

American Mathematical Society MATHSCINET MATHEMATICAL REVIEWS

But did you know about these?

LENS.ORG Solving The Problem Of Problem Solving™

BASE

CORE

Nexis Uni®

SEMANTIC SCHOLAR A free, AI-powered research tool for scientific literature

Dimensions

[You can find a list with links & descriptions here: Information tools]

Look at category *Discovery* and *Search* for these tools and more.

And did you know there is a whole world of other types of search tools out there?

Local Citation Network

ResearchRabbit

Citation Tree

citationchaser

CONNECTED PAPERS

OPEN KNOWLEDGE MAPS

Elicit

PURE suggest

consensus

Citation Graph

Litmaps

Inciteful

scite_

citation gecko

SCISPACE

Search Tools – extra recommendations

You can find the main recommendations on the UT library site. The ones listed on this slide are (mostly) new, free, different, and, most importantly, very useful!

www.semanticscholar.com

SEMANTIC SCHOLAR
A free, AI-powered research tool for scientific literature

Good alternative for google scholar. Semantic search, no Booleans and such, good recommendations in my experience

www.lens.org

LENS.ORG
Solving The Problem Of Problem Solving™

Great free alternative for scopus/WoS, with a more modern interface, better search tools, and often better results

Two slides with more info on these tools can be found by clicking this:



inciteful.xyz

Inciteful

Awesome way to quickly get lots of extremely relevant recommendations based on your current library.
With the Zotero plugin it's literally 1 click to use!

https://fabian-beck.github.io/pure-suggest/

PURE suggest

Combine your library with keywords to quickly find the most relevant connected papers

elicit.org

Elicit

Uses GPT-3.5 to find relevant papers and quickly extract crucial info

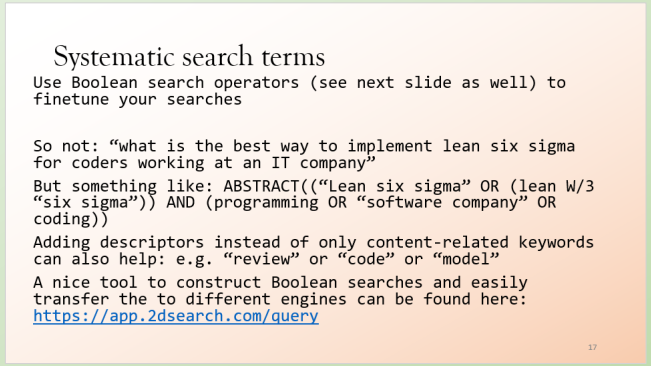
www.researchrabbit.ai

ResearchRabbit

Like Inciteful and PURE suggest combined and on steroids, dive into a deep rabbit hole of recommendations and citations

Search Tools – Boolean search engines

All the ‘classic’ search engines support Boolean search operators for systematic searching. The library guides focus on these types of search engines. I included a couple of slides with tips, videos, and guides as well.



Systematic search terms

Use Boolean search operators (see next slide as well) to finetune your searches

So not: “what is the best way to implement lean six sigma for coders working at an IT company”

But something like: ABSTRACT((“Lean six sigma” OR (lean W/3 “six sigma”)) AND (programming OR “software company” OR coding))

Adding descriptors instead of only content-related keywords can also help: e.g. “review” or “code” or “model”

A nice tool to construct Boolean searches and easily transfer the to different engines can be found here: https://app.2dsearch.com/query

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Doing the work

Example on how to use the tools and interpret the results

Recent Papers by the Top 100 Authors

	adamic_adar	published_year	num_cited_by
Analysis of the Library Novelty Search Service Scheme Based on ICT Technology Siyu Xia *Proceedings of the 2022 3rd International Conference on Big Data and Informatization Education (ICBDIE 2022)*	1.703185	2022	0
Application Analysis of Large Data and Net of Things Technology in the Construction of Intelligent Library Xin Zhou *2022 11th International Conference on Software and Information Engineering*	1.442695	2022	0
The Application of Big Data Technology in University Library Management Service Jianan Zhong *Lecture notes on data engineering and communications technologies*	0.910239	2021	0
Hotspot Mining in the Field of Library and Information Science under the Environment of Big Data Shujie Jiang *Journal of Environmental and Public*	0.910239	2022	0

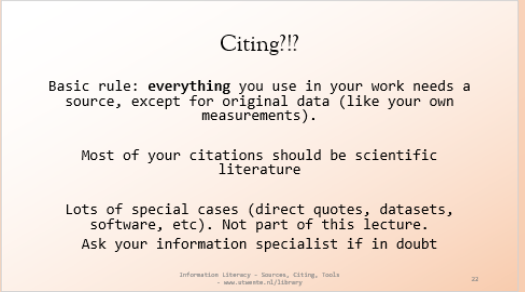
The Most Important Recent Papers

	num_cited_by	page_rank	published_year
Radio frequency identification and internet of things-based smart library management system Phani Gannamraju, ... L. S. Kumar *International Journal of Networking and Virtual Organisations*	2	0.29025	2021
Opportunities for academic smart libraries in the Caribbean Adrian St. Patrick Duncan *Library Hi Tech News*	2	0.2265	2021
Adoption of Smart Technologies in University Libraries of Pakistan: A Qualitative Review Abid Hussain, Pervaiz Ahmad	1	0.21375	2021
Development of an IoT-enabled Smart Library System for a University Campus Yash Gupta Sungkur, ... Soulakshmee D. Nagowah *Journal of Telecommunication, Electronic and Computer Engineering*	1	0.210563	2021
Exploration on Smart Library Construction in 5G Era Lei Zhang *Journal of physics*	2	0.203125	2021

includes data collection and processing, how to judge whether events are triggered, how the system

Read, cite, write

- Obvious: don’t read non-relevant work
- Cite properly. **Use a reference manager!!** Check which style to use with your supervisor.
- Official rules for plagiarism are quite easy: **everything** that isn’t **directly produced by you** needs a source. More info on hidden slides (click) →
- Always immediately cite your source, or **at least** put down a placeholder [?].



The Complete Process

(I’m just going to say “search for stuff with good tools and select relevant results, repeat if you want more, oh and be smart about it”)
(but with more words)

The Complete Process

You should know the answer to these questions:

What do I want to know? Which sources will answer my question?

Where do I find this information? Which tools and databases can I use?

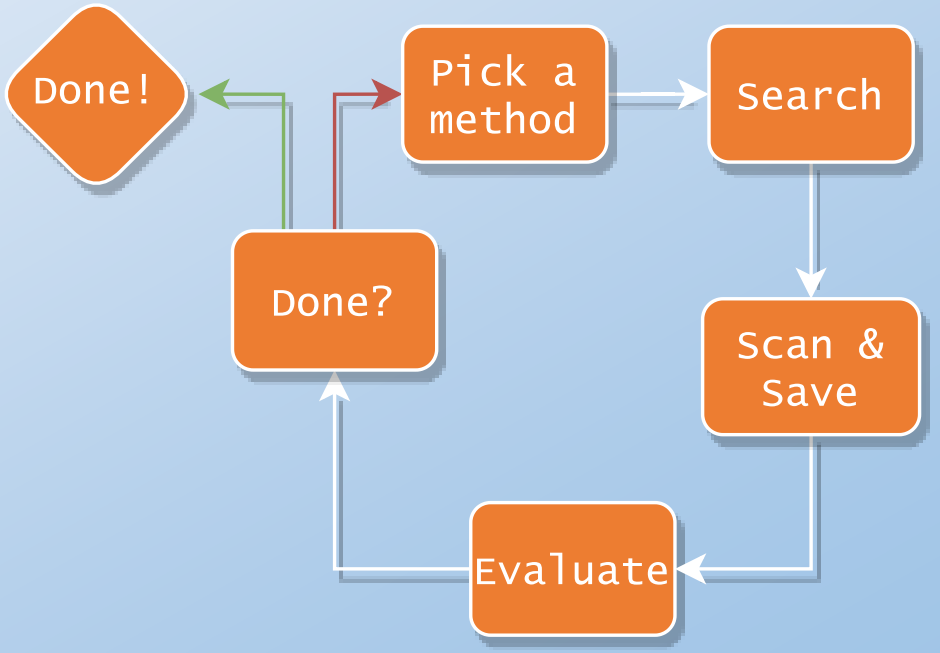
And how do I use them?

Can I trust & use what I found? Why?

How do I cite my sources?

Search Cycle

1. Pick a search method, e.g., keywords, semantic search, knowledge graph, citation map
2. Start searching
3. Find relevant papers in the results: e.g., scan title, abstract; use filters; explore citation network, surfaced metadata
4. Evaluate results, read the papers to determine if you're satisfied
5. If:
   A. Done? Stop searching (for now)
   B. Not done? Start again; including new sources



Research Cycle

Searching

UT Library

Starting point for all your research – don't forget the guides and FindUT!

Aggregated databases

Broader search. Examples: Scopus, Web of science, Lens, Semantic Scholar

Specific databases

Full-text search and more detail. Examples: IEEE, ScienceDirect, NexisUni

Managing

Access to articles

Plugins & links: Lean Library, Unpaywall, UT network, Ezproxy, Open access

Saving articles

Reference manager: Zotero, Mendeley, Endnote

Install browser plugin for your reference manager to make citing really easy and effortless

Read, write, repeat

Notes

take notes and annotate papers, identify keywords, concepts

Cite

Integrate your reference manager with your text editor for easy and correct citations

Discover more

Use your current bibliography and insight to kickstart the next round of searching.

Go back to the start, but now also use tools like inciteful, pure suggest, etc to help you alongside keyword searches

Wrapping up

What now?

1. Install software: reference manager, lean library, plugins, ……
2. Write an outline of your research question/model, decide on boundaries (time, subjects,…)
3. Write down (roughly) what you need to know (see slide 25)
4. Start your search cycle (slide 27/28)
5. Don’t spend too long searching, it’s better to stop earlier to process the info. You can always go back and search for more

Need help?

Can’t find your sources? Trouble with a tool? Reference manager isn’t working for you? No access to a critical article?

s.mok@utwente.nl

Just send me an email and I’ll help you! It helps if you provide some details in the question, and we can also plan a meeting if you need some more help.

That's it!

If there's time left: Questions are welcome of course.

I wish you all the best of luck with your research projects!